

Agenda

- Modern Data Centers
- Design Goals of Data Center Networks
- (Old) Data Center Networks
- **Fat-Tree Network Topology**
- Virtual Layer 2 (VL2) Networks

Case Study: Fat-Tree

- **[DC-Fat-Tree]** M. Al-Fares, A. Loukissas and A. Vahdat, "A Scalable, Commodity Data Center Network Architecture," In. *Proc of ACM SIGCOMM'08*, Aug 2008.
- To address the limitations of the old architecture
 - single point of failure
 - over subscription of links higher up in the topology ← trade-offs between cost and providing
- A new Fat-tree topology to increase bi-section bandwidth ← any two servers have **multiple** paths!
- **Novel addressing, forwarding/routing**

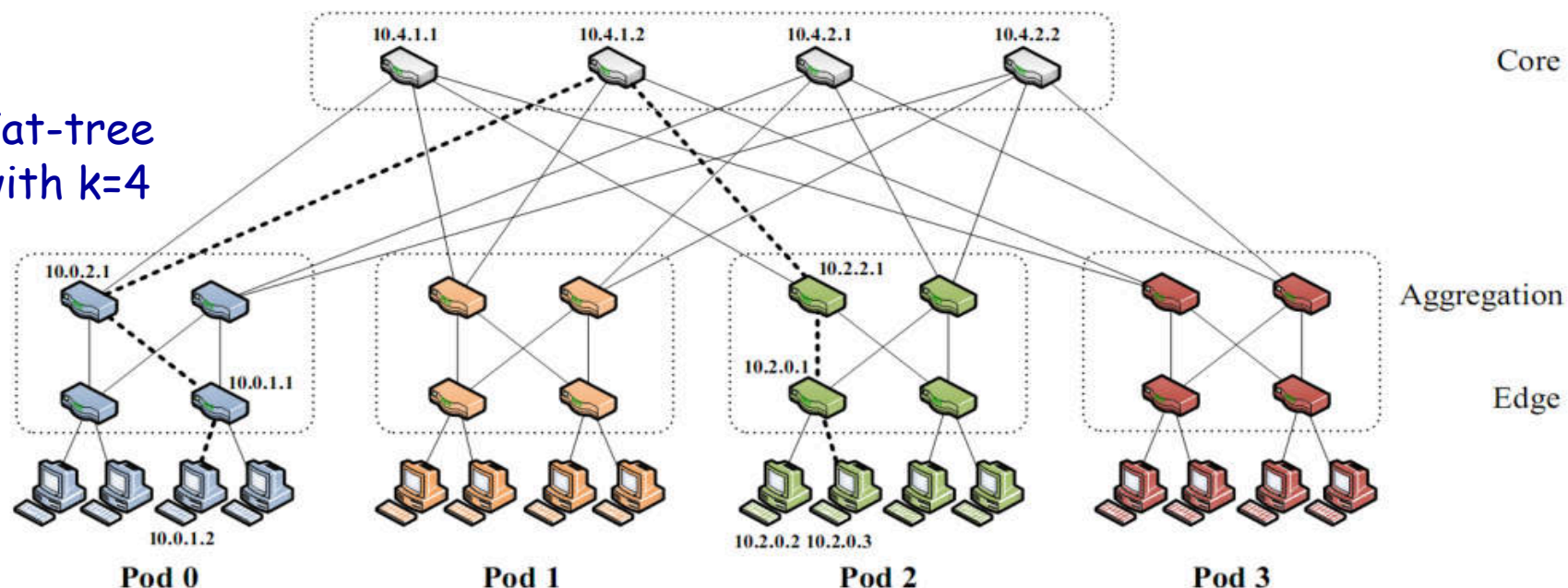
Fat-Tree's Design Goals

- ▶ Allows host communication at line speed
 - ▶ no matter where they are located!
- ▶ Backwards compatible with existing infrastructure
 - ▶ no changes in application & support of layer 2 (Ethernet)
- ▶ Cost effective
 - ▶ cheap infrastructure
 - ▶ low power consumption & heat emission

K-nary Fat-Tree Architecture

- Fat-Tree: a special type of Clos Networks (after C. Clos)
 - each pod consists of $(k/2)^2$ servers & 2 layers of $k/2$ k -port switches
 - each edge switch connects to $k/2$ servers & $k/2$ aggr. switches
 - each aggr. switch connects to $k/2$ edge & $k/2$ core switches
 - $(k/2)^2$ core switches: each connects to k pods

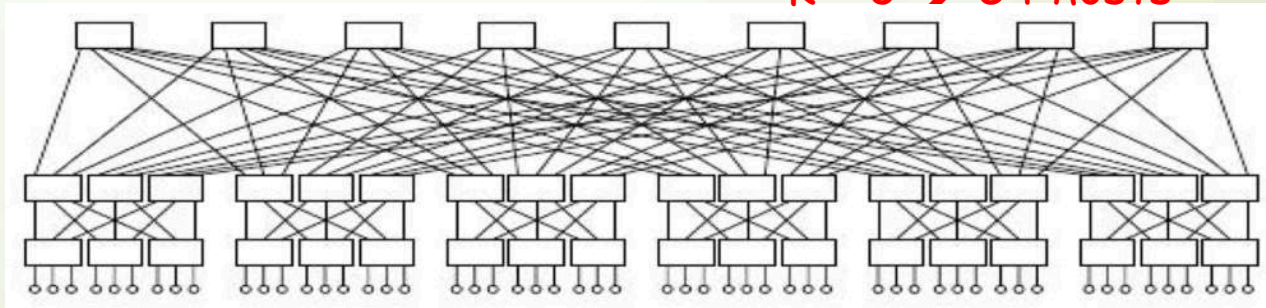
Fat-tree
with $k=4$



Pros of Fat-Tree Architecture

- Why Fat-Tree?
 - Fat tree has **identical bandwidth** at any bisections
 - Each layer has the **same** aggregated bandwidth
- Can be built using cheap devices with **uniform capacity**
 - Each port supports same speed as end host
 - All devices can transmit at line speed **if packets are distributed uniform along available paths**
- Great scalability: k -port switch supports $k^3/4$ servers

$K = 3 \rightarrow 54$ hosts



How About Routing/Switching?

➤ Ethernet switching (layer 2)

- ✓ Fixed IP addresses and auto-configuration (plug & play)
- ✓ Seamless mobility, migration, and failover
- ✗ Broadcast limits scale (ARP)
- ✗ No multipath (Spanning Tree Protocol)

➤ IP routing (layer 3)

- ✓ Scalability through hierarchical addressing
- ✓ Multipath routing through equal-cost multipath
- ✗ Can't migrate w/o changing IP address
- ✗ Complex configuration

Fat-Tree Goes with Layer-3 Routing

- Layer 2 switch algorithm: data plane flooding!
- Layer 3 IP routing:
 - shortest path IP routing will typically use only one path despite the path diversity in the topology
 - if using equal-cost multi-path routing at each switch independently and blindly, packet re-ordering may occur; further load may not necessarily be well-balanced
 - Aside: control plane flooding!
- OK, what is their neat proposal?

Addressing and Routing in Fat-Tree

- **Enforce** a special (IP) addressing scheme
 - unused.PodNumber.switchnumber.Endhost
 - Allows host attached to same switch to route only through switch
 - Allows inter-pod traffic to stay within pod
- Use two level look-ups to distribute traffic and maintain packet ordering ← **compatible to TCAM lookup!**

- First level is prefix lookup
 - used to route **down the topology to servers**
- Second level is a suffix lookup
 - used to route **up towards core**
- **maintain packet ordering by using same ports for same server**

Prefix	Output port
10.2.0.0/24	0
10.2.1.0/24	1
0.0.0.0/0	

Suffix	Output port
0.0.0.2/8	2
0.0.0.3/8	3

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Case Study: VL2

- **[VL2]** A. Greenberg, et al., "VL2: A Scalable and Flexible Data Center Network", In. Proc of ACM SIGCOMM'09, Aug 2009.
- A virtual (logical) layer 2 for the whole network
 - employs a 3-level Clos topology (full-mesh in top-2 levels) with non-uniform switch capacities
- Separate identity and location
 - application and location addresses
 - employs a directory service for name resolution
 - but needs direct host participation (agents on servers)
- Explicitly accounts for DC traffic matrix dynamics
 - Valiant load-balancing (VLB) → using randomization to cope with volatility